ENTERPRISE STUDENT RECORD AND ACADEMIC MANAGEMENT PLATFORM

Unit 1 Mini Project – Product Engineering & Design Thinking

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Project Basis

This mini project is prepared using the supplied Unit-1 mini-project brief and the supplied training guide. The brief requires business understanding, Design Thinking, IBM Design Thinking, stakeholder analysis, personas, requirements, product vision, MVP, roadmap, architecture, project planning and repository structure.

1. Business Requirements Document (BRD)

The platform is designed for a higher-education institution to manage student records and academic information in one centralized system. The solution covers admissions, course registration, attendance, examinations, results and academic reporting.

Business Problems

• Manual records can be duplicated, lost or become outdated.

• Students need quick access to attendance and academic information.

• Faculty need efficient attendance and result management.

• Academic administrators need centralized records and reliable reports.

• Examination staff need consistent result generation and record tracking.

Business Objectives

• Centralize student and academic records.

• Reduce manual data-entry errors.

• Improve transparency of attendance and results.

• Generate academic reports quickly.

• Provide secure role-based access.

2. Design Thinking Report

The supplied project brief requires the five stages: Empathize, Define, Ideate, Prototype and Test.

Empathize

Identify the experiences of students, faculty members, academic administrators, examination-cell staff and IT support personnel. Focus on delays, repeated manual work, difficulty finding records and lack of real-time information.

Define

Problem Statement: Students and academic staff face delays and errors because academic information is often managed across disconnected or manual processes. A centralized platform is needed to make records, attendance, examination data and reports easier to manage.

Ideate

• Student profile and registration module

• Attendance dashboard

• Examination and result module

• Academic report generation

• Role-based dashboards

• Search and notification features

Prototype

A low-fidelity prototype should include Login, Student Dashboard, Faculty Dashboard, Attendance, Examination/Results, Reports and Administrator Dashboard screens.

Test

Validate the prototype with representative students, faculty and administrators. Check navigation clarity, task completion, correctness of displayed academic information and usefulness of reports. Record feedback and revise the prototype iteratively.

3. IBM Design Thinking

The project applies the three principles specified in the brief: Focus on User Outcomes, Restless Reinvention and Diverse Empowered Teams.

• Focus on User Outcomes: measure whether users can complete academic tasks faster and with fewer errors.

• Restless Reinvention: continuously improve workflows based on feedback and usage observations.

• Diverse Empowered Teams: involve students, faculty, administrators, developers, UX designers, QA and DevOps/IT roles.

IBM Loop: Observe → Reflect → Make

Observe user interactions and operational problems → Reflect on feedback and measurable outcomes → Make changes to the design, requirements or implementation → repeat the cycle.

4. Stakeholder Analysis

Stakeholder	Role / Need
Students	Register, view profile, attendance, examination information and academic reports.
Faculty Members	Manage attendance, academic records and marks/results.
Academic Administrators	Manage student records, courses and institutional reports.
Examination Cell	Maintain examination information and generate results.
IT Support	Maintain system availability, security, backups and technical support.

5. User Personas

Persona	Goals	Challenges	Expected Features
Student	Access records quickly; monitor attendance and results.	Manual processes; delayed information.	Student dashboard, attendance, results, reports.
Faculty	Record attendance and marks efficiently.	Repeated entry; difficulty tracking records.	Attendance module, marks entry, class reports.
Administrator	Maintain accurate institutional records and reports.	Distributed records; manual reporting.	Admin dashboard, search, reporting, user management.

6. Functional Requirements Specification (FRS)

ID	Requirement
FR-01	Secure login and role-based access for Student, Faculty, Administrator, Examination Cell and IT roles.
FR-02	Student registration and profile management.
FR-03	Course/subject registration and academic record management.
FR-04	Attendance entry, calculation and viewing.
FR-05	Examination information and marks/result management.
FR-06	Academic report generation.
FR-07	Search and retrieve student academic records.
FR-08	Administrative dashboard and record management.

7. Non-Functional Requirements Specification (NFRS)

Category	Requirement
Security	Authentication, authorization, secure password handling and protected academic data.
Usability	Simple navigation and clear dashboards for students and staff.
Reliability	Records should be stored consistently and remain available during normal operations.
Performance	Common searches and dashboard operations should respond quickly.
Scalability	Architecture should support growth in students, courses and records.
Maintainability	Use modular code, documentation, version control and clear folder structure.

8. Product Vision

Vision: To provide a centralized, secure and user-friendly academic management platform that simplifies student record handling, attendance, examinations and academic reporting for higher-education institutions.

Mission: Reduce manual administrative effort and provide accurate academic information to students and staff through a single digital platform.

Success Criteria

• Students can access their academic information easily.

• Faculty can update attendance and academic records efficiently.

• Administrators can generate reports from centralized data.

• Role-based access protects academic information.

• User feedback shows improved usability over manual processes.

9. Minimum Viable Product (MVP)

The supplied mini-project brief defines the first release around student registration, record management, attendance and academic report generation.

MVP Feature	Purpose
Student Registration	Create and maintain student records.
Record Management	Store student, course and academic information.
Attendance	Enter, calculate and display attendance.
Academic Reports	Generate basic student academic reports.
Login & RBAC	Provide controlled access by user role.

10. Product Roadmap

Phase	Planned Work
Phase 1	Business understanding, stakeholder study and requirements.
Phase 2	Design Thinking, personas and low-fidelity prototype.
Phase 3	Agile backlog, sprint planning and project setup.
Phase 4	Core application development: database, backend and frontend.
Phase 5	Testing and DevOps automation / CI-CD.
Phase 6	Deployment, monitoring, feedback and continuous improvement.

11. Initial System Architecture

A layered architecture is proposed so that the user interface, application services and data storage remain separated.

```
Users
  |
  v
Web / Mobile User Interface
  |
  v
Application / API Layer
  |-- Authentication & RBAC
  |-- Student Records
  |-- Attendance
  |-- Examination & Results
  |-- Reporting
  |
  v
Database Layer
  |-- Students
  |-- Courses
  |-- Attendance
  |-- Examinations
  |-- Results
  |
  v
Backup / Monitoring / Deployment Services
```

12. Project Repository Structure

```
student-academic-management-platform/
■■■ .github/
■   ■■■ workflows/
■■■ app/
■   ■■■ modules/
■   ■   ■■■ students/
■   ■   ■■■ attendance/
■   ■   ■■■ examinations/
■   ■   ■■■ reports/
■   ■■■ database/
■   ■■■ tests/
■■■ docs/
■■■ frontend/
■■■ backend/
■■■ docker/
■■■ monitoring/
■■■ README.md
■■■ .gitignore
```

Git should be initialized at the repository root, followed by an initial commit containing the project structure and documentation.

13. Project Charter

Item	Details
Project Name	Enterprise Student Record and Academic Management Platform
Purpose	Centralize student and academic information and reduce manual administrative effort.
Primary Users	Students, Faculty, Academic Administrators, Examination Cell, IT Support
MVP Scope	Registration, record management, attendance, reports and role-based access.
Key Deliverables	BRD, product vision, stakeholder analysis, personas, FRS, NFRS, Design Thinking report, MVP, roadmap, architectu
Methodology	Design Thinking followed by Agile product development and DevOps practices.
Major Risks	Incorrect requirements, data-security issues, poor user adoption and integration complexity.

Item	Details
Mitigation	User validation, iterative prototypes, role-based security, testing and continuous feedback.

14. Product Discovery Documentation

Discovery should capture stakeholder interviews/observations, user pain points, problem statements, assumptions, proposed solutions, prototype feedback and decisions. The discovery record should be updated throughout the Observe → Reflect → Make cycle.

Key Discovery Questions

• What academic tasks take the most time today?

• Which records are difficult to locate or update?

• What information should students see immediately?

• What reports do faculty and administrators need most often?

• What security and privacy controls are required?

15. Conclusion

The mini project converts the academic-management problem into a product-engineering plan before coding begins. It follows the supplied brief by covering business understanding, Design Thinking, IBM Design Thinking, stakeholders, personas, requirements, product vision, MVP, roadmap, architecture, project charter and repository planning. The resulting specification provides a practical foundation for Agile planning and later implementation.

Source Basis

This report is based on the uploaded Unit-1 mini-project brief and the uploaded DevOps Unit-1 training/project guide. The brief specifies the required deliverables, including BRD, Product Vision, Stakeholder Analysis, Personas, FRS/NFRS, Design Thinking, IBM Design Thinking, MVP, roadmap, architecture, project charter, repository structure and product discovery documentation.

The training guide supplies the product-engineering framing, Design Thinking stages, IBM Loop, stakeholder/persona examples, requirements-engineering concepts, MVP thinking, roadmap approach and repository-structure concept.